How the AIS models score

Updated comparison of RPF agreement, MolProbity geometry and ProteinEBM energy for SR215.

12,104
Models with all three scores

1,200
New classic AFSample models

12
AIS sampling methods

Median score by AIS method

AIS method	Models	MolProbity	Clashscore	Energy t=0.10	RPF DP	Recall	Precision
AFCluster	1,000	3.75	64.30	3,974.1	0.492	0.890	0.868
AFSample classic	1,200	3.51	60.92	4,051.4	0.737	0.953	0.908
AFSample2	1,000	3.86	53.06	4,159.7	0.686	0.935	0.902
AFSample3	1,000	1.65	8.54	3,894.3	0.775	0.958	0.921
AlphaFlow MD	993	2.34	32.37	4,150.4	0.674	0.931	0.900
AlphaFlow PDB	990	3.37	80.04	4,228.3	0.661	0.937	0.889
BioEMU1	992	3.82	99.15	4,172.0	0.559	0.931	0.850
BioEMU2	1,000	2.60	17.09	3,998.7	0.661	0.935	0.890
Boltz2 Exact	1,000	1.77	18.88	3,785.9	0.695	0.952	0.891
ESMFlow PDB	929	3.33	82.73	4,215.1	0.631	0.934	0.878
OpenDDE Exact	1,000	1.48	8.09	3,820.5	0.775	0.959	0.920
SPEACHAF	1,000	3.47	39.12	4,116.0	0.736	0.945	0.914

Classic AFSample median DP was 0.737. Highest median DP was 0.775 (AFSample3, OpenDDE Exact).

OpenDDE Exact had the lowest median MolProbity (1.48). Boltz2 Exact had the lowest median energy (3,785.9).

Coverage All 1,200 classic AFSample models scored after the 20-model proton completion. The original 10,904 scores were reused. The 96 unscored models from the original AIS cohort remain outside this comparison.

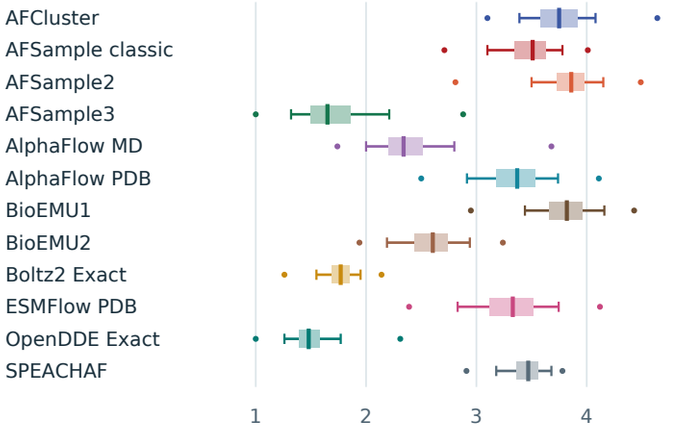
Scores RPF uses the TR NOESY lists. MolProbity measures stereochemical quality. Higher DP and lower MolProbity are favorable. ProteinEBM Model 6 energy is a learned score in arbitrary units. Main comparisons use t=0.10, with lower energy ranked first.

Score distributions by AIS method

Boxes show the middle 50%, center marks show medians, whiskers show the 5th to 95th percentiles, and dots show extrema.

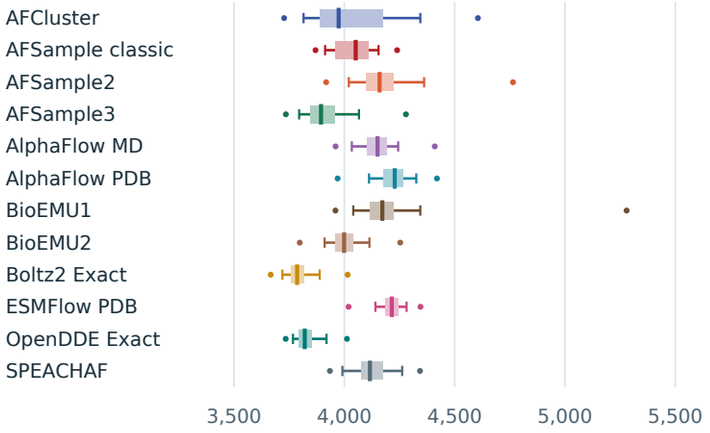
MolProbity score

lower is better



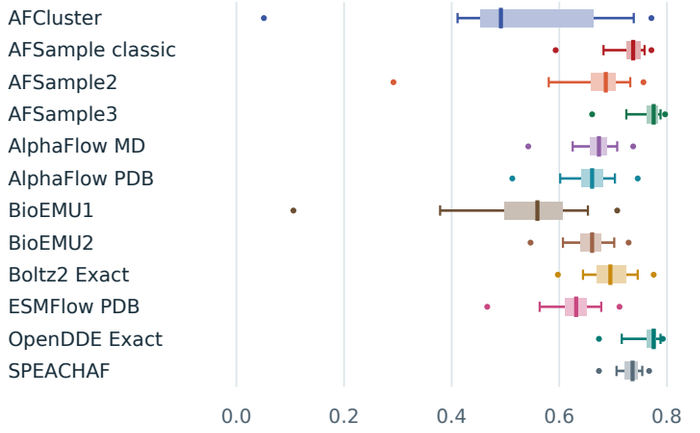
ProteinEBM t=0.10

lower ranks first



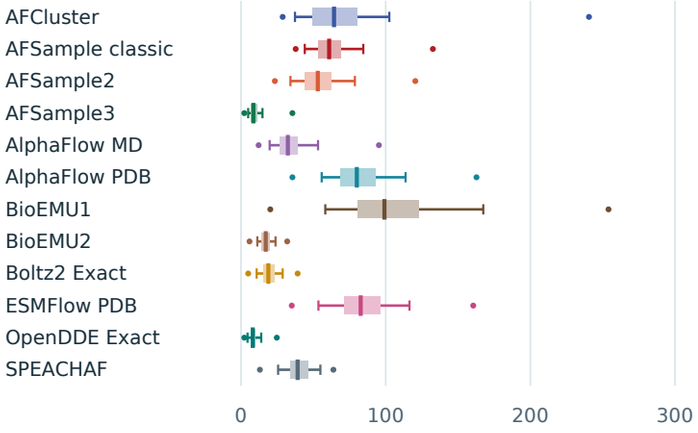
RPF DP

higher is better



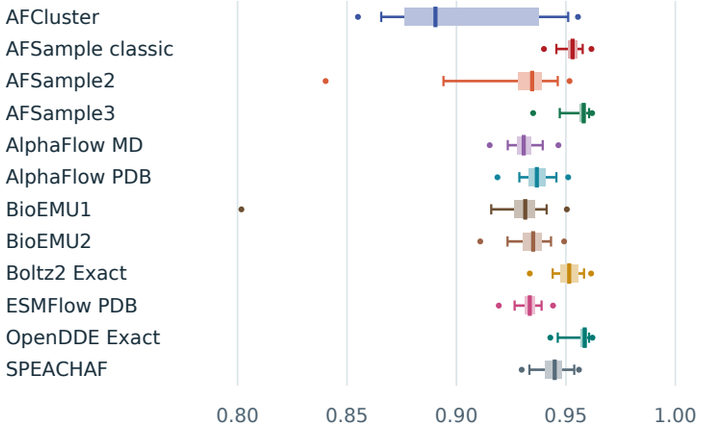
Clashscore

lower is better



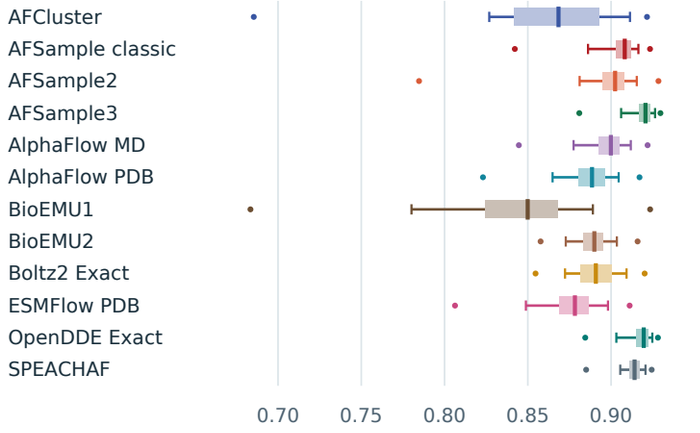
RPF recall

higher is better



RPF precision

higher is better



All 12,104 models are included in every panel. Scales are shared across AIS methods within each panel.

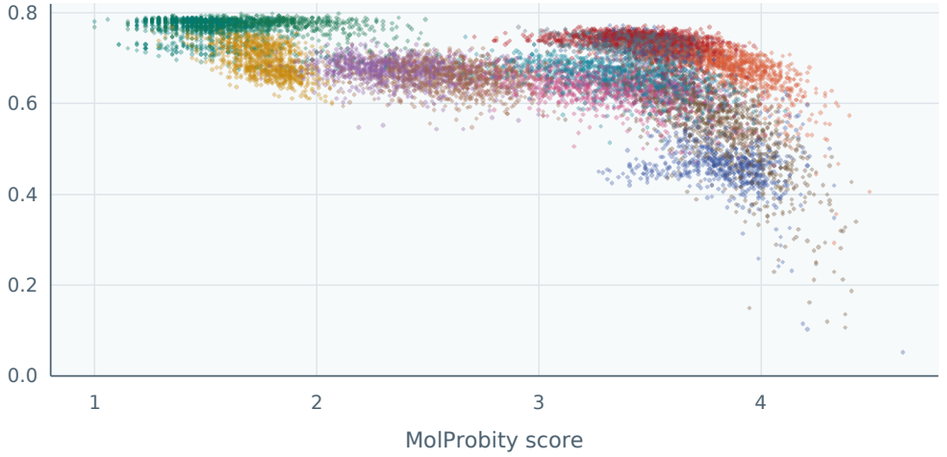
Agreement among the three main scores

Each dot represents one model. All 12,104 matched models contribute to each pairwise comparison.

Pooled rank correlations combine differences within methods and differences between methods.

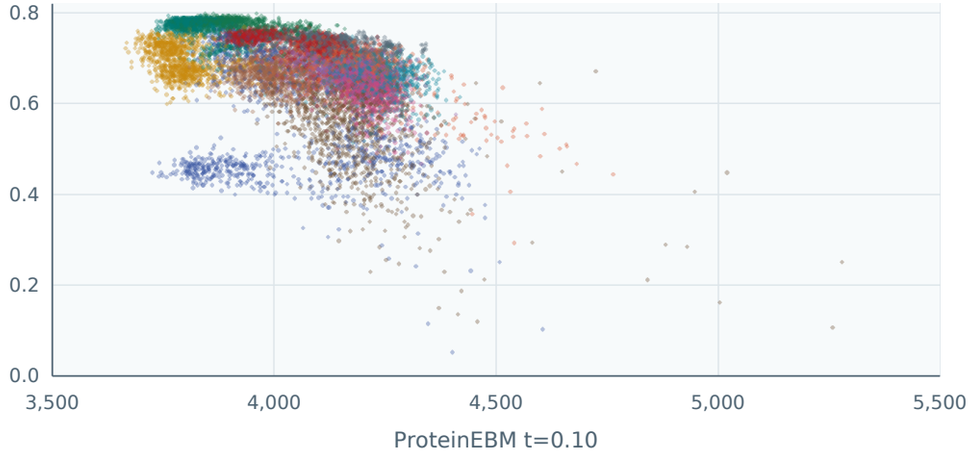
RPF DP vs MolProbity score

rho -0.55



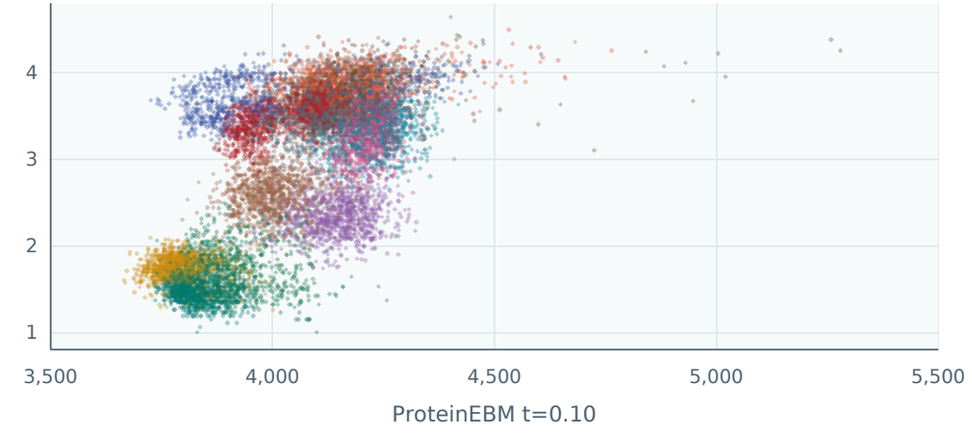
RPF DP vs ProteinEBM t=0.10

rho -0.53



MolProbity score vs ProteinEBM t=0.10

rho +0.60



AFCluster · AFSample classic · AFSample2 · AFSample3 · AlphaFlow MD · AlphaFlow PDB · BioEMU1 · BioEMU2 · Boltz2 Exact · ESMFlow PDB · OpenDDE Exact · SPEACHAF

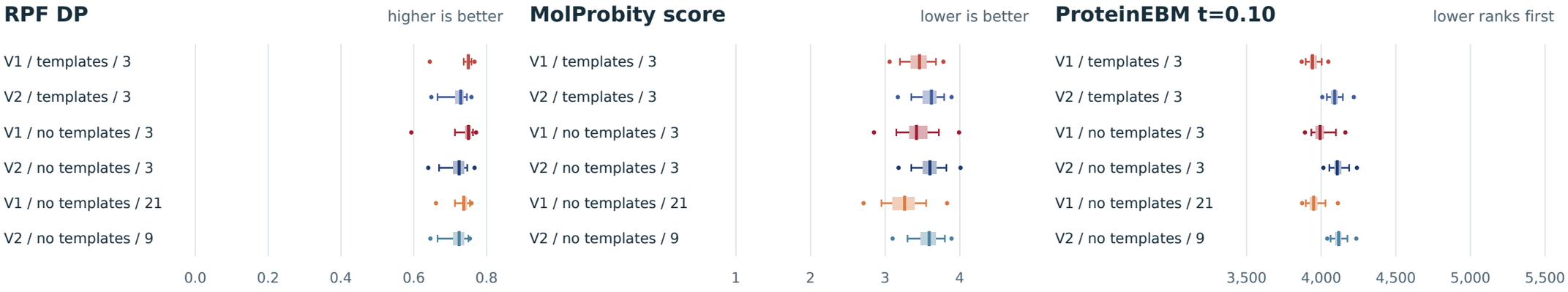
Spearman rho by AIS method

AIS method	MP vs DP	Energy vs DP	Energy vs MP
All	-0.55	-0.53	+0.60
AFCluster	-0.60	-0.13	+0.48
AFSample classic	-0.43	-0.58	+0.58
AFSample2	-0.69	-0.36	+0.21
AFSample3	-0.12	-0.31	+0.06
AlphaFlow MD	-0.27	-0.20	+0.10
AlphaFlow PDB	-0.47	-0.25	+0.10
BioEMU1	-0.71	-0.32	+0.25
BioEMU2	-0.28	-0.27	+0.12
Boltz2 Exact	-0.53	-0.32	+0.15
ESMFlow PDB	-0.50	-0.21	+0.23
OpenDDE Exact	-0.16	-0.23	-0.15
SPEACHAF	-0.27	-0.52	+0.22

Classic AFSample across six conditions

Each condition contains 200 models. Labels give checkpoint family, template setting and maximum recycle count.

Condition	Scored	Amide fixes	Median DP	DP ≥ 0.75	MolProbity	Clashscore	Energy t=0.10
V1 / templates / 3	200	7	0.750	114	3.46	63.40	3,941.2
V2 / templates / 3	200	0	0.729	9	3.62	63.85	4,089.9
V1 / no templates / 3	200	10	0.750	128	3.42	64.53	3,992.7
V2 / no templates / 3	200	0	0.725	10	3.60	57.55	4,107.8
V1 / no templates / 21	200	3	0.737	30	3.26	57.55	3,949.0
V2 / no templates / 9	200	0	0.725	14	3.59	59.80	4,116.9



305 of 1,200 models reached DP ≥ 0.75. The V1 three-recycle conditions supplied 242 of those models.

The initial run held 20 models with missing amide H atoms at distorted peptide links. Native fragment completion restored all 20.

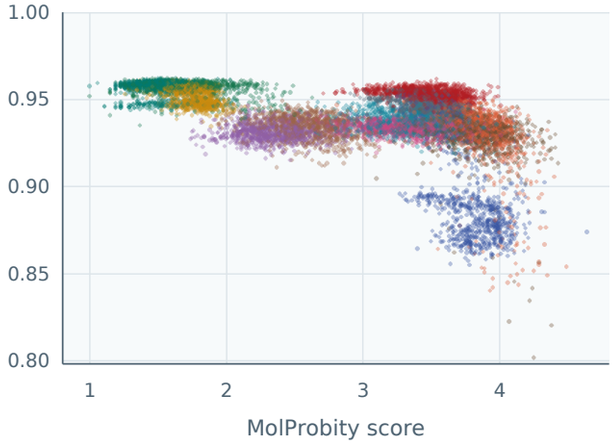
The 21 added amide H atoms preserved the original heavy coordinates and existing atom records. The peptide distortions are retained.

Condition comparisons describe the scored models and combine checkpoint, template and recycle settings.

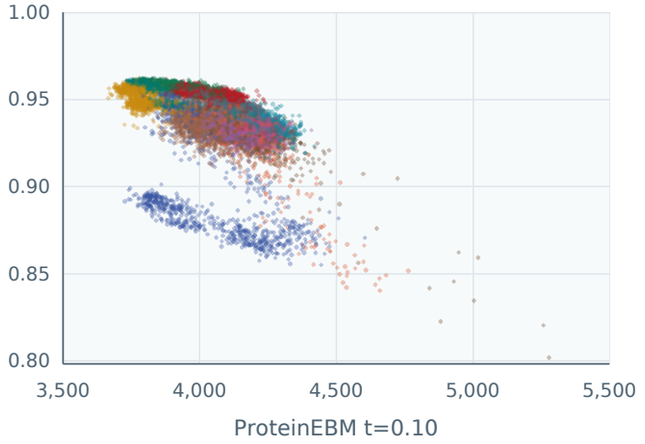
Recall, precision and clashscore

Each panel uses all 12,104 models. Rho denotes pooled Spearman rank correlation.

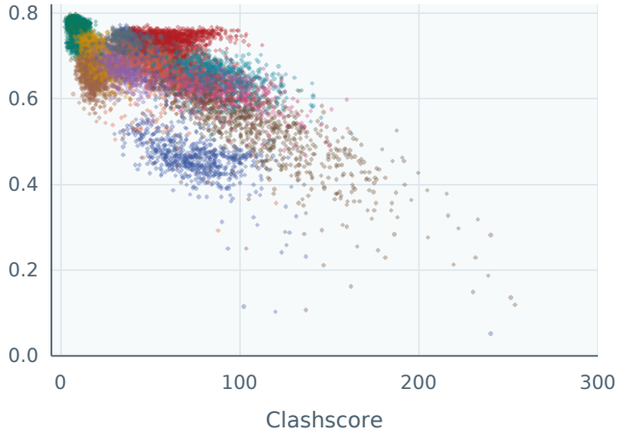
RPF recall vs MolProbity score — rho -0.56



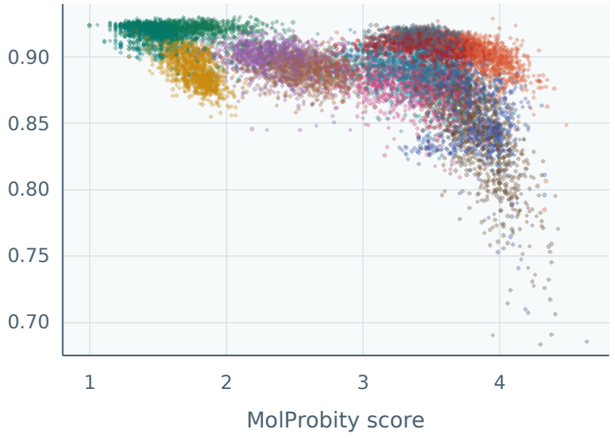
RPF recall vs ProteinEBM t=0.10 — rho -0.67



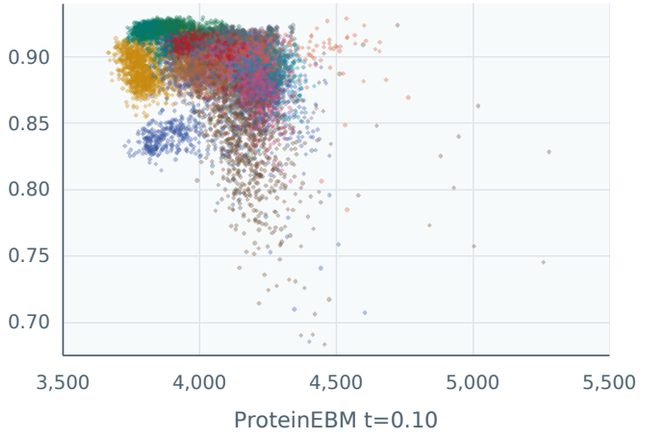
RPF DP vs Clashscore — rho -0.65



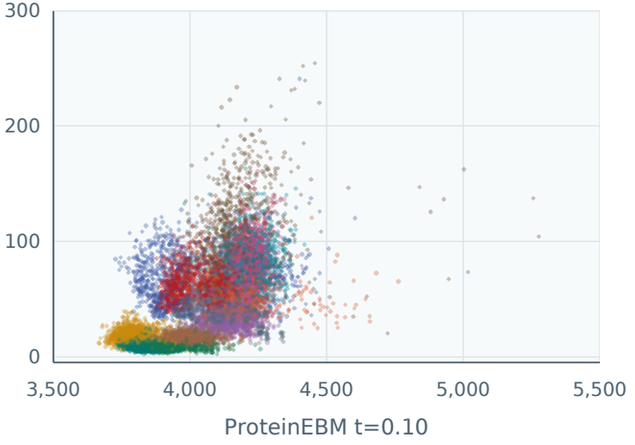
RPF precision vs MolProbity score — rho -0.47



RPF precision vs ProteinEBM t=0.10 — rho -0.37



Clashscore vs ProteinEBM t=0.10 — rho +0.63



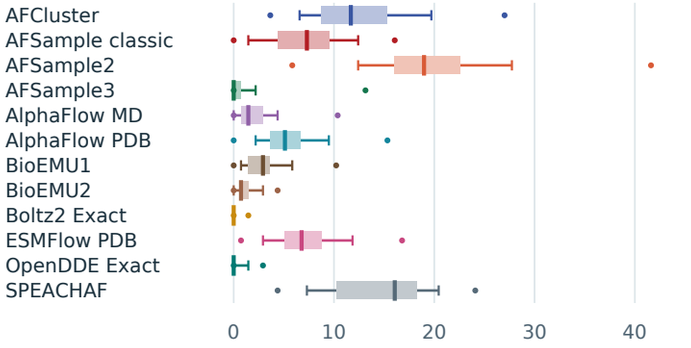
- AFCluster
- AFSample classic
- AFSample2
- AFSample3
- AlphaFlow MD
- AlphaFlow PDB
- BioEMU1
- BioEMU2
- Boltz2 Exact
- ESMFlow PDB
- OpenDDE Exact
- SPEACHAF

Geometry details and energy sensitivity

RPF and MolProbity use validated protonated queries. ProteinEBM scores the original structures at 16 conditioning levels.

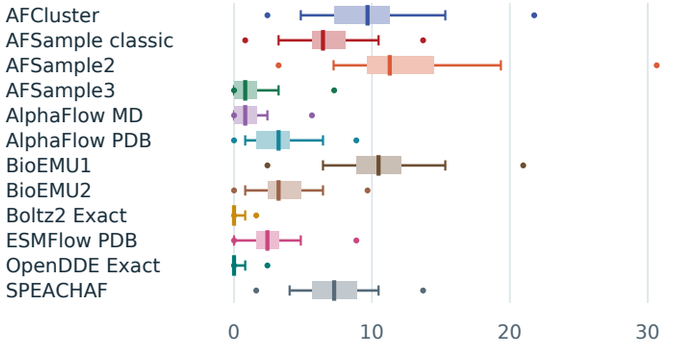
Ramachandran outliers (%)

lower is better



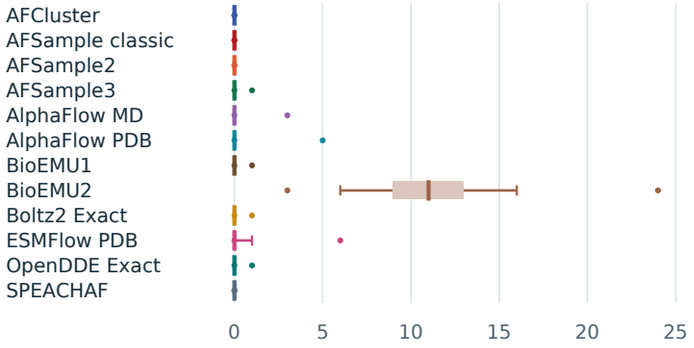
Rotamer outliers (%)

lower is better



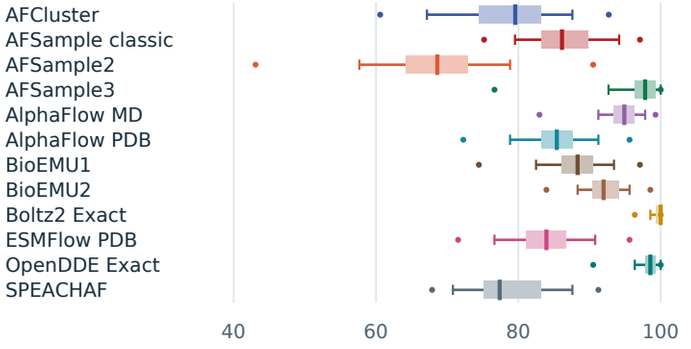
C-beta deviations

lower is better



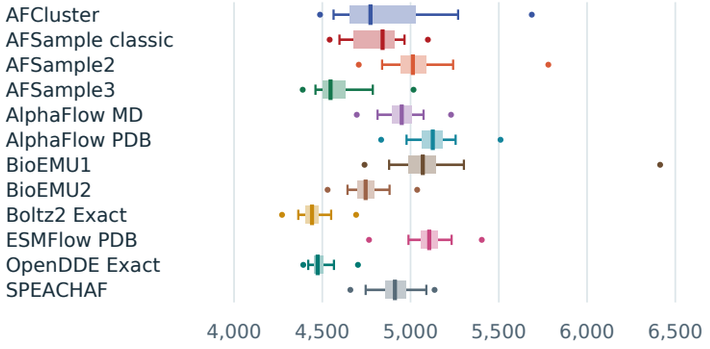
Ramachandran favored (%)

higher is better



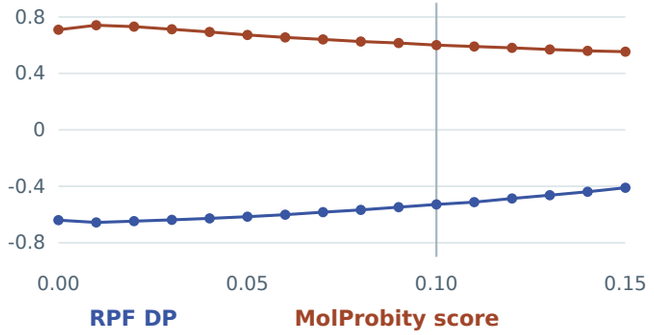
Mean ProteinEBM energy

mean over 16 levels



ProteinEBM across conditioning levels

Pooled rank correlation vs score



Preparation and sources

The earlier cohort includes 985 side-chain rebuilds and 159 local geometry repairs. ProteinEBM used the original models for both groups.

All 1,200 classic AFSample queries retain their original heavy coordinates. Their ProteinEBM runs use the same Model 6 expert checkpoint.

Classic AFSample model intake 23601, original scoring audit 23605, completion audit 24185, and ProteinEBM audit 24197.

RPF scorer 8920fc9e15dd / TR original control / Phenix 2.2.1-6174 / ProteinEBM Model 6 checkpoint 2d3045666c89.

The condition levels are ProteinEBM inputs. Energy values retain their original arbitrary units across the full 139-residue construct.